PURUSHOTTAM DIWAKAR



ARIANA AGARWAL, 16; AVYANA MEHTA, 16
Class 12 students at Jayshree Periwal International School
Jaipur

A Smart Filter

It was during a field trip to the villages near Jaipur that the trio of Ariana Agarwal, Avyana Mehta and their schoolmate, Vivaan Chhawchharia, noticed the villagers drinking contaminated water. They realised that while affordable methods to remove certain impurities were available, there were hardly any low-cost solutions for eliminating microplastics, an environmental and public health threat. Over time, microplastics can build up in organs like the liver and blood, and trigger cell swelling or hormone disruption. Researching for remedies, the trio learnt about the adhesive properties of tamarind seed powder and began exploring its potential in isolating microplastics. An opportunity to work under mentors at IIT Guwahati helped sharpen their focus.

After numerous trials, they developed a formulation by combining tamarind seed powder with iron particles. Natural polysaccharides extracted from the seeds bind to suspended microplastics, while embedded magnetite nanoparticles allow the resulting flocs (fluffy clumps of bacteria/fungal filaments found in water) to be removed using a handheld magnet. "For Plas-Stick, a major turning point was getting our dosage right. We had to go through several iterations to find the right balance. The breakthrough was the moment we realised that our idea could genuinely work beyond the lab," says school head girl Ariana, who had also worked on GRAINOS, a moisture sensor device that analyses wheat grains to predict their likely future market value for farmers.

For Avyana, a pivotal moment was starting Plastic2Build, her youth-led non-profit. "What began as an attempt to address the plastic waste crisis gradually grew into something much larger," she says. ■

—Rohit Parihar

WHAT MAKES THEM SHEROES

Along with classmate Vivaan Chhawchharia, Ariana and Avyana developed Plas-Stick, a biomagnetic coagulant derived from tamarind seeds that helps remove microplastics from water. Won them the Global Winner title at The Earth Prize

TAKEAWAYS

"What makes our innovation special is that it uses something simple and indigenous to address a global problem"
—Ariana

"We started with a simple premise: can something we see as waste be part of the solution to a much bigger environmental problem?"
—Avyana